Points to Clarify

1. **DEMAND AND INVENTORY CALCULATION DETAILS:**

   i) **3 Month Sales:**
   - The distribution of SKUs among stores for new products was based on the 7-day rolling demand share per store for each SKU. These demands were rounded to integers.

   ii) **ROP, EOQ, Safety Stock:**

   - **Average Daily Demand:** Calculated as the average daily demand of each SKU in a store over the three months.

   - **Pct_zero_sales_day:** Represents the percentage of days with zero sales for each SKU-store combination.

   - **Safety stock** = Z* √ (lead time) * (standard deviation of daily demand)[1]
   - **Reorder point**= ((Average daily demand) * (Lead time)) + Safety Stock [2]

   - **EOQ**: The formula used for EOQ is not the classic formula, but the order-level EOQ for perishables:

     a) EOQ = (Average Daily Demand) * (Shelf life - Average lead time)[3]

   iii) **Pareto Analysis:** All parameters are for the particular SKU ACROSS ALL STORES

2. **Lost Sales Not Reduced to Zero**

i) The model does not always eliminate lost sales. This is primarily due to certain SKU-store combinations starting with very low stock levels. Even when replenishment orders are placed on time, sales losses can occur during supplier lead time.

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[1] Source: "Understanding Safety Stock and Mastering Its Equations" by Peter L. King, MIT

[2] Source: "Operations Management" by Jay Heizer and Barry Render

[3] Source: Inventory and Production Management in Supply Chains (Fourth Edition) by Edward A. Silver, David F. Pyke, Rein Peterson

ii) This is a limitation of the competition's rules on starting inventory, not the model design.

3. **Reason for Choosing March 20 as Start Date:** The start date was chosen to ensure sufficient historical data for the Holt-Winters method, allowing more accurate capture of seasonal demand patterns.

4. **Forecast Methodology**

i) Holt-Winters Seasonal Forecasting:

- Holt-Winters was employed for SKU-store pairs exhibiting periodic spikes, supporting accurate day-wise predictions by capturing trend and seasonality in demand.
- This approach improves forecast precision and enables timely and quantity-optimized order placement.[4]

5. **Additional Clarification Points**

i) Initial Inventory Conditions

- The model assumes starting inventory values as given by the dataset, which may be low for some SKU-store pairs. This impacts initial lost sales and is a limitation external to model logic.

ii) Data Integrity and Demand Patterns

- Demand data quality is critical. Missing or incorrect data can affect forecast accuracy. The model relies heavily on historical demand patterns to generate forecasts.

iii) Model Scope and Competition Rules

- The model strictly adheres to competition constraints, including the simulation window, SKU

iv) Presenter Notes

- Presenters' notes are added as a separate document as not all points could have been added to the presentation

[4] Source: Hyndman, R.J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.), OTexts, Section 7.3 "Holt-Winters' seasonal method".